CHA-4.D.1 A function defined as an integral represents an accumulation of a rate of change.
CHA-4.D.2 The definite integral of the rate of change of a quantity over an interval gives the net change of that quantity over that interval.
CHA-4.E.1 The definite integral can be used to express information about accumulation and net change in many applied contexts.

8-3 Using Accumulation Functions and Definite Integrals in Applied Contexts

Recall in unit 6 we discussed integrals. Now we're going to discuss applications of integrals:

$$\int \text{rate of change} = \text{Net Change}$$

of ___ of ___

Remember the Fundamental Theorem of Calculus:

$$\int_a^b f'(x)dx = f(b) - f(a)$$

For contextual situations, you can interpret this as saying

> "If you integrate the rate of change of a function over an interval from a to b, you will determine the amount of change in that function between a and b."

By transposing the terms in the equation above, another version looks like this:

$$f(b) = f(a) + \int_a^b f'(x)dx$$

For contextual situations, you can interpret this as saying

> "You will determine the value of the function at b, if you start with an initial value at a and add the amount of change that has taken place over the interval from a to b."

Example 1: The initial population of a town is 5500 people, and the population is changing at the rate of $P'(t)$ people per year. Write, but do not evaluate, an integral expression that will determine the population of the town after 10 years.

$t=0$ $t=10$

$P(0)=5500$

$P(10)=?$

$$P(10) = P(0) + \int_0^{10} P'(t)dt$$

$$= 5500 + \int_0^{10} P'(t)dt$$

net change in Pop.

Translated: The population at 10 years is the initial population plus the net change in the population during the 10 years.

AP Note: It is imperative that you write the equation, not just the answer in FRQs to receive credit!!! You **do not** need to rewrite the entire function being used for the integrand if you use the name given to the function instead. This is particularly helpful for complex functions because it avoids copy errors.

Example 2: There are 30 gallons of pollutant in a lake at $t = 0$. If the number of gallons, $P(t)$, of the pollutant changes at the rate of $P'(t) = 1 - 3e^{-0.3\sqrt{t}}$ gallons per day, where t is measured in days. Determine the amount of pollutant in the lake after 12 days.

$P(0) = 30$ gal. $P(12) = ?$

$$P(12) = P(0) + \underbrace{\int_0^{12} P'(t)\,dt}_{\text{net change in Pollutants 0 to 12 days}}$$

$$= 30 + \int_0^{12} P'(t)\,dt$$

$$= 23.4220286\ldots$$

The amt. of pollutant in the lake after 12 days is 23.422 gal

Example 3: Water is leaking from a tank at a rate of $\sqrt{t+1}$ gallons per minute. Ten minutes after the leak is discovered, there are 60 gallons of water in the tank. How much water was in the tank when the leak was initially discovered?

$t=0$ Amt? $t=10$ min Amt $=60$ gal.

$$\text{Amt at } t=0 = 60 + \underbrace{\int_0^{10} \sqrt{t+1}\,dt}_{\text{how much leaked out from 0 to 10min}}$$

Example 4: A pie is removed from an oven and, after sitting for 10 minutes, is at a temperature of 220 degrees Fahrenheit. The Pie cools at a rate of $R(x) = 3\ln(x^2)$ degrees Fahrenheit per minute. What is the expected temperature of the pie 15 minutes after it was removed from the oven?

$T(10) = 220°F$ $\quad T(15) = ?$ $\quad$ Let $T(x)$ = temp. of the pie

$$T(15) = T(10) - \underbrace{\int_{10}^{15} R(x)\,dx}_{\text{How many degrees the Pie Cooled}}$$

$$= 220 - \int_{10}^{15} R(x)\,dx$$

$$= 144.4305875\ldots$$

The expected temp. of the Pie 15 min after it has been removed is 144.431°F

The following problems very closely resemble a type of problem often found on the AP Calculus exam in the Free Response section. They are commonly referred to as "Rate In/Rate Out Problems."

Example 5: A store is having a 5-hour sale. The rate at which shoppers enter the store t hours after the sale begins is modeled by the function $E(t)$, which is measured as shoppers per hour. The rate at which shoppers leave the store t hours after the sale begins is modeled by the function $L(t)$. When the sale starts, there are already 70 shoppers in the store. Write an expression to find out how many shoppers are in the store at the end of the 5 hour sale.

$$\text{Amt of shoppers in the store at } t=5 \text{ hrs} = 70 + \underbrace{\int_0^5 E(t)\,dt}_{\text{\# of shoppers that entered}} - \underbrace{\int_0^5 L(t)\,dt}_{\text{\# of shoppers that Exited}}$$

$$= 70 + \underbrace{\int_0^5 \left[E(t) - L(t)\right] dt}_{\text{net change in num. of shoppers}}$$

Example 6: Throughout the day, people have entered a craft fair at the rate of $E(t) = 30t^2 + 16t$ people per hour and have exited at a rate given by $L(t) = 12t - 3$ people per hour. If t is measured in hours since the fair opened, and there were 200 people at the fair at the end of the first hour, how many people should be at the fair 3 hours after it opened?

200 people $t=1$ people? when $t=3$

Num of People at the fair at $t=3$ $= 200 + \int_1^3 [E(t) - L(t)]\,dt$

$= 482$ people

482 people should beat the fair 3 hrs after it opened.

Example 7: 2017 AB 2

When a certain grocery store opens, it has 50 pounds of bananas on a display table. Customers remove bananas from the display table at a rate modeled by

$$f(t) = 10 + (0.8t)\sin\left(\frac{t^3}{100}\right) \quad \text{for } 0 \le t \le 12$$

where $f(t)$ is measured in pounds per hour and t is the number of hours after the store opened. After the store has been open for three hours, store employees add bananas to the display table at a rate modeled by

$$g(t) = 3 + 2.4\ln(t^2 + 2t) \quad \text{for } 3 < t \le 12$$

where $g(t)$ is measured in pounds per hour and t is the number of hours after the store opened.

(a) How many pounds of bananas are removed from the display table during the first 2 hours the store is open?

$$\int_0^2 f(t)\ dt = 20.051175\ldots \qquad 2: \begin{cases} 1: \text{integral} \\ 1: \text{answer} \end{cases}$$

20.051 pounds of bananas are removed from the display table during the first 2 hours the store is open.

(b) Find $f'(7)$. Using correct units, explain the meaning of $f'(7)$ in the context of the problem.

$f'(7) = -8.120$ (or -8.119)

AP EXAM HINT:
On the calculator FRQ section, DO NOT compute derivatives by hand! Use your graphing calculator to find $f'(7)$.

Remember when explaining the meaning of a derivative in context, use this format (from Sec 4-1)

Interpreting a Derivative in Context
Interpret $f'(c)$ given $f(c)$

The _____ is _____ at a rate of _____ at
$f(t)$ in context — increasing/decreasing — $f'(c)$ with units

$t =$ _____
time with units

The rate at which number of bananas is removed from the table is decreasing at a rate of 8.120 (or 8.119) pounds per hour per hour when $t = 7$ hours.

Alternate answer: After the store has been open 7 hours, the rate at which bananas are being removed from the display table is decreasing by 8.120 (or 8.119) pounds per hour per hour

Example 7: 2017 AB 2

When a certain grocery store opens, it has 50 pounds of bananas on a display table. Customers remove bananas from the display table at a rate modeled by

$$f(t) = 10 + (0.8t)\sin\left(\frac{t^3}{100}\right) \quad \text{for } 0 \le t \le 12$$

Rate removed

where $f(t)$ is measured in pounds per hour and t is the number of hours after the store opened. After the store has been open for three hours, store employees add bananas to the display table at a rate modeled by

$$g(t) = 3 + 2.4\ln(t^2 + 2t) \quad \text{for } 3 < t \le 12$$

rate added

where $g(t)$ is measured in pounds per hour and t is the number of hours after the store opened.

(c) Is the number of pounds of bananas on the display table increasing or decreasing at time $t = 5$? Give a reason for your answer.

$f(5) = 13.7959384$
$g(5) = 11.5328353$

$2: \begin{cases} 1: \text{considers } f(5) \text{ and } g(5) \\ 1: \text{answer with reason} \end{cases}$

The number of pounds of bananas on the display table is decreasing at time $t = 5$ because the rate they are being removed is greater than the rate they are being added.

Rate the num of bananas is changing $= g(t) - f(t)$

Alternate solution:
$g(5) - f(5) = -2.263103 < 0$

Because $g(5) - f(5) < 0$, the number of pounds of bananas on the display table is decreasing at time $t = 5$.

(d) How many pounds of bananas are on the display table at time $t = 8$?

$$50 + \int_3^8 g(t)\,dt - \int_0^8 f(t)\,dt = 23.347396\ldots$$

$3: \begin{cases} 2: \text{integrals} \\ 1: \text{answer} \end{cases}$

23.347 pounds of bananas are on the display table at time $t = 8$